Conda: The Soul of Anaconda

Conda, which is included in Anaconda and Miniconda, is an open source package management system and environment management system for installing multiple versions of software packages and their dependencies, and switching easily between them. It is multiplatform, working on Linux, OS X and Windows, and was created for Python programs but can package and distribute any software.

Remember 'Anaconda', that horror movie with the tag line "You can't scream if you can't breathe."? Well, a few years ago, Python had an encounter with Anaconda, and today, it acts as the backbone of Anaconda. Confused? I am now referring to Anaconda, the Python distribution that acts both as a package manager and an environment manager. But before we talk a bit more about *this* Anaconda, here's a brief introduction to package managers.

Package management

Some applications cannot stand alone. They need the support of other applications to work. The applications that need to be installed for the proper working of an application are considered its dependencies, e.g., IPython needs *python-decorator* and *python-simplegeneric* to be installed in a system to work properly.

You can install packages either manually or by using some package managers. If you install a package manually, that package alone will be installed. Its dependencies should, therefore, be installed separately. As the number of dependencies increases, it becomes difficult to install all the packages manually. A package manager deals with this problem. According to Wikipedia, a package manager is a collection of software tools that automates installation, updation, configuration and removal of software in a consistent manner. Thus, a package manager resolves all the dependencies of a given software.

The following are some of the package managers available for Linux distributions.

1. *dpkg:* A low level package management system, it uses the Debian repository to install packages that come in the *.deb* format. All the dependencies of the package to be installed will be contained within the *.deb* file. The command…

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dpkg -i <package-name>
```

…can be used to install a package.